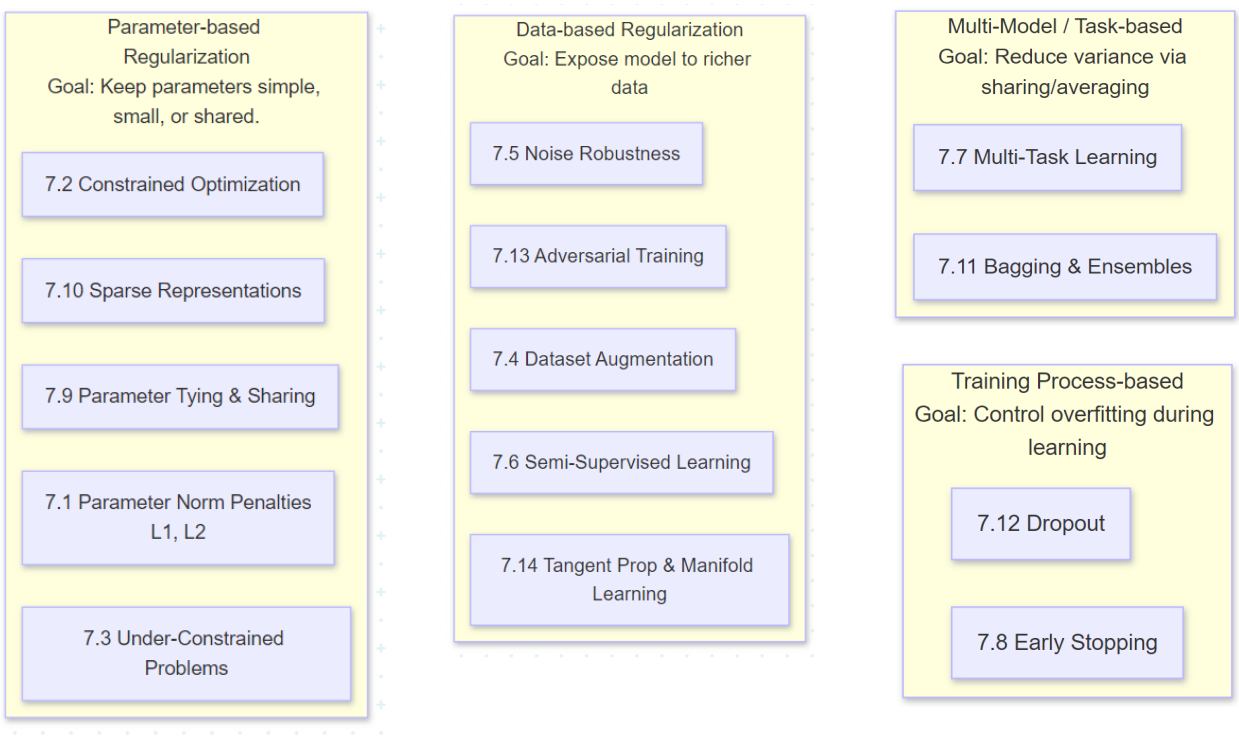
UNIT – II

Regularization for Deep Learning

(14 regularization methods)

Regularization = ***techniques to prevent overfitting and improve generalization in deep learning.***

Grouping of Techniques



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1. Parameter-based Regularization

(penalizing or controlling model weights)

Goal: Keep parameters simple, small, or shared.

- 7.1 Parameter Norm Penalties (L1, L2)
- 7.2 Norm Penalties as Constrained Optimization
- 7.3 Regularization in Under-Constrained Problems
- 7.9 Parameter Tying & Sharing
- 7.10 Sparse Representations

2. Data-based Regularization

(making better use of data)

Goal: Expose the model to richer or trickier data → generalize better.

- 7.4 Dataset Augmentation
- 7.5 Noise Robustness
- 7.6 Semi-Supervised Learning
- 7.13 Adversarial Training
- 7.14 Tangent Distance / Tangent Prop / Manifold Tangent Classifier

3. Training Process-based Regularization

(adjusting how training is done)

Goal: Control overfitting during learning.

- 7.8 Early Stopping
- 7.12 Dropout

4. Multi-Model / Task-based Regularization

(using multiple tasks or models)

Goal: Share information across tasks or average across models → reduces variance.

- 7.7 Multi-Task Learning
- 7.11 Bagging & Ensembles

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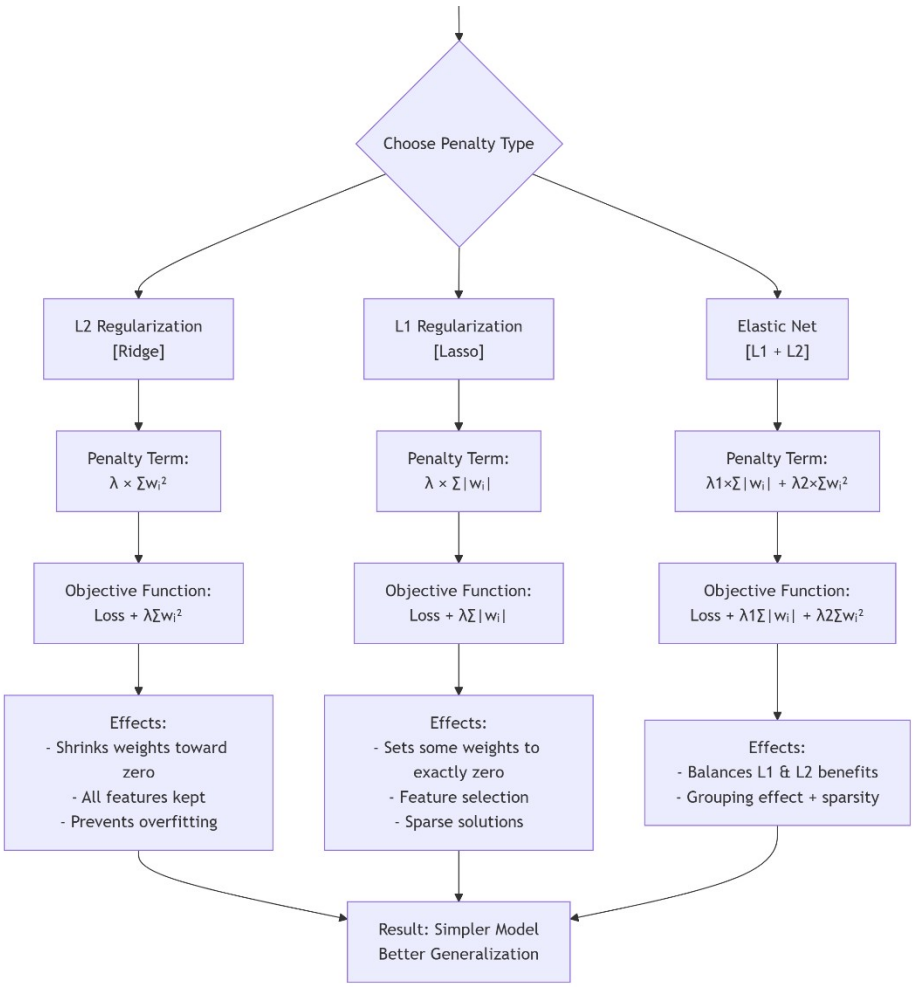
7.1 Parameter Norm Penalties

Definition: Add a penalty on large weights (L1, L2 norms).

Example: L2 (Ridge) regression shrinks weights towards zero.

Short Explanation: Prevents model from relying too much on a few parameters; makes it simpler and less likely to overfit.

Input Data → Model Training → + Penalty Term → Regularized Objective → Optimize → Simpler Model



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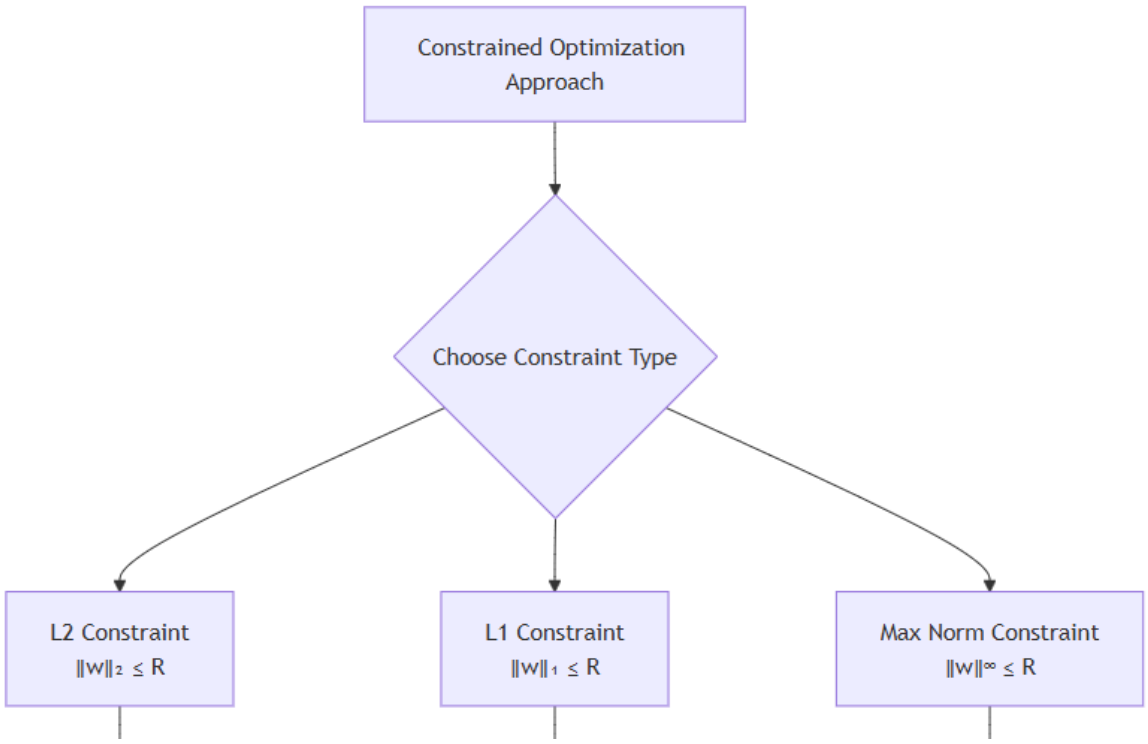
7.2 Norm Penalties as Constrained Optimization

- **Definition**: Instead of penalizing weights, constrain them directly (e.g., “keep weights within a ball of radius R”).
- **Example**: "Weights must have length ≤ 5."
- **Short Explanation**: Equivalent to putting a boundary around model parameters so they don’t explode.



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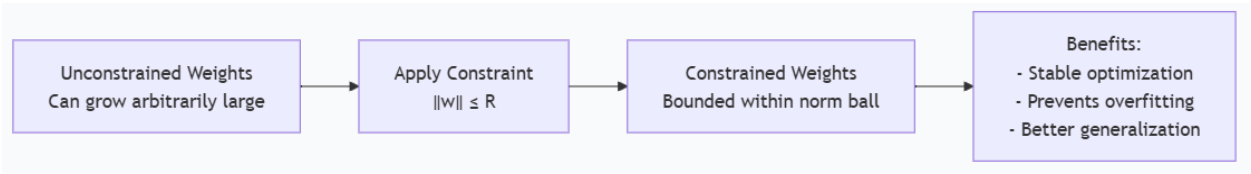

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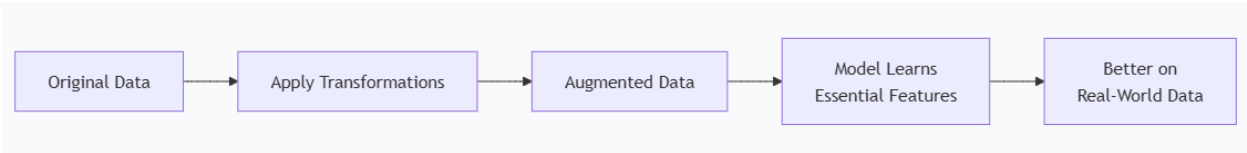
7.3 Regularization and Under-Constrained Problems

- **Definition**: Regularization helps when many possible solutions fit data (too few constraints).
- **Example**: Fitting a line to only 2 points → infinite lines possible; regularization picks the simplest.
- **Short Explanation**: Regularization guides model to a unique, stable solution.

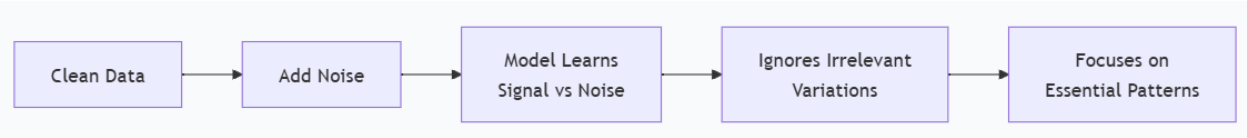
7.4 Dataset Augmentation

- **Definition**: Create new training data by transforming existing samples.
- **Example**: Rotate or flip images in image classification.
- **Short Explanation**: Makes model robust to variations; teaches it general patterns.



7.5 Noise Robustness

- **Definition**: Add noise to inputs, outputs, or hidden units to prevent overfitting.
- **Example**: Add small random pixel noise to MNIST digits.
- **Short Explanation**: Model learns to ignore irrelevant variations → more general.



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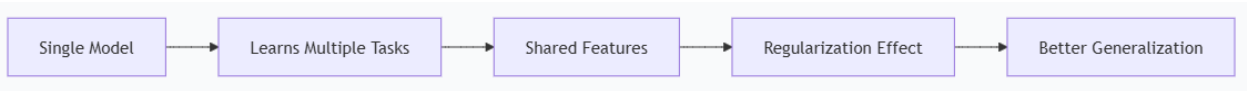
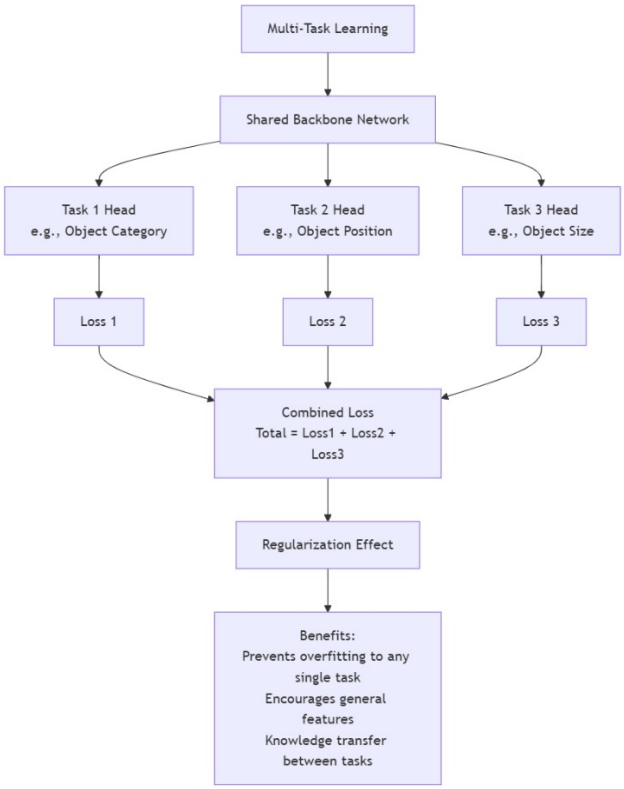
7.6 Semi-Supervised Learning

- **Definition**: Use both labeled and unlabeled data for training.
- **Example**: Only 100 labeled images + 10,000 unlabeled images.
- **Short Explanation**: Helps when labeled data is scarce; regularization encourages consistency on unlabeled data.

7.7 Multi-Task Learning

- **Definition**: Train a model on multiple tasks simultaneously.
- **Example**: One network predicts both "object category" and "object position."
- **Short Explanation**: Sharing knowledge between tasks prevents overfitting to just one.





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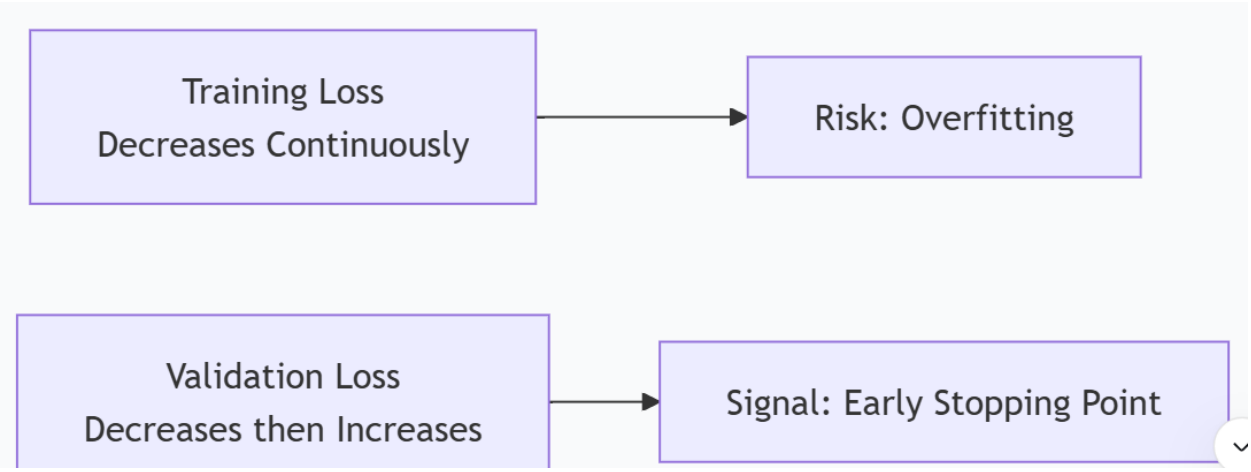
7.8 Early Stopping

- **Definition**: Stop training when validation error stops improving.
- **Example**: Training accuracy increases, but validation accuracy drops → stop there.
- **Short Explanation**: Prevents overfitting by not letting the model "memorize" data.



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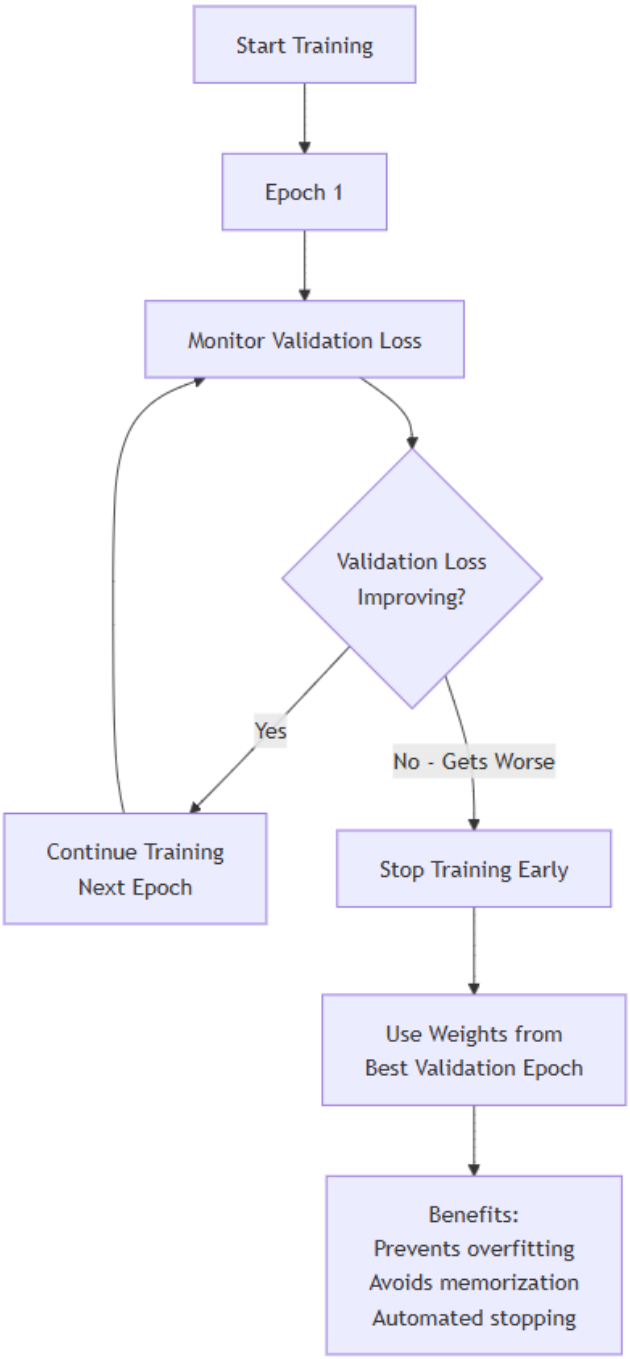

7.9 Parameter Tying and Parameter Sharing

- **Definition**: Force some parameters to be the same or reused.
- **Example**: Convolutional Neural Networks (CNNs) share filters across image positions.
- **Short Explanation**: Reduces number of parameters; increases generalization.

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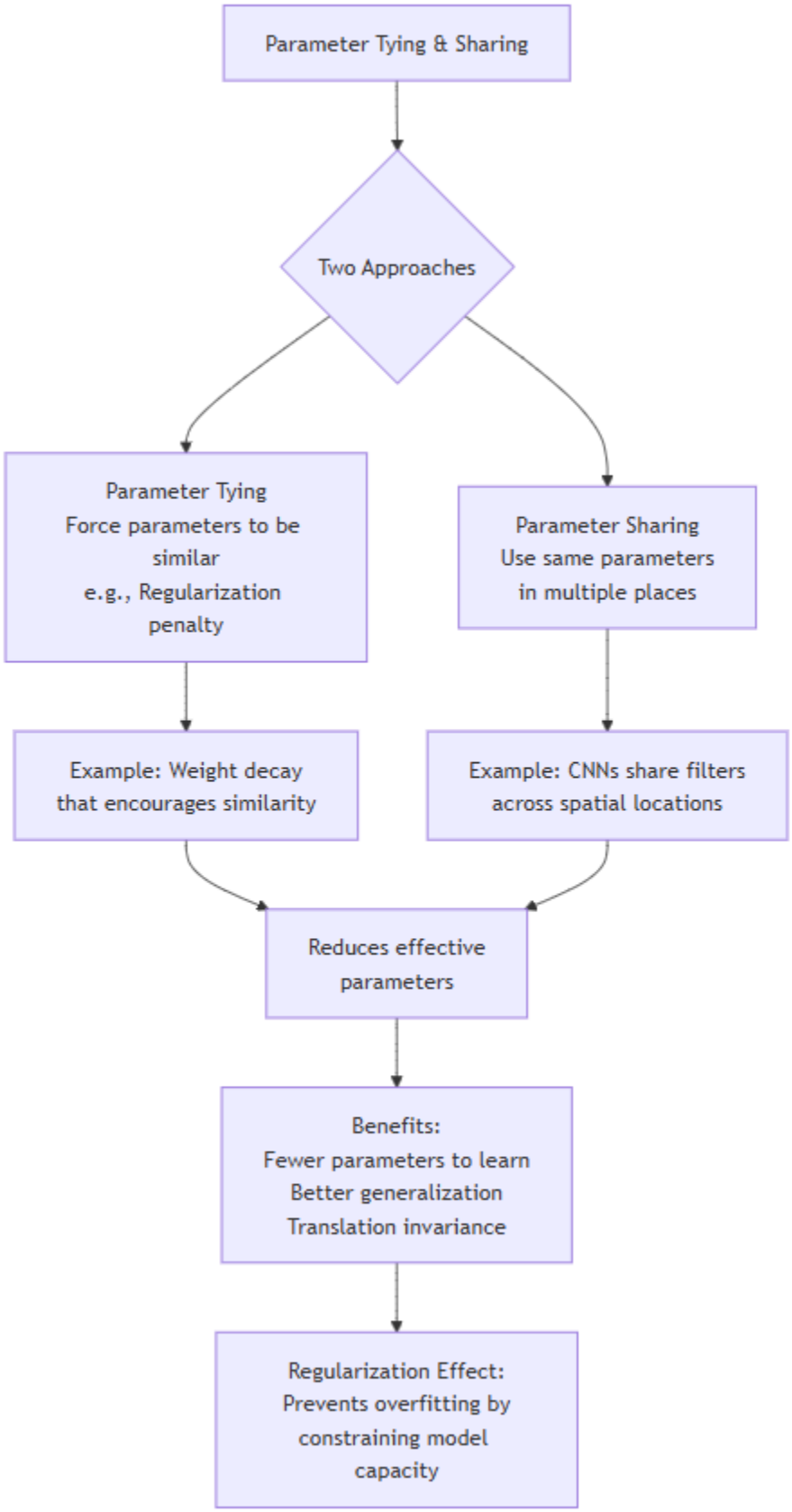

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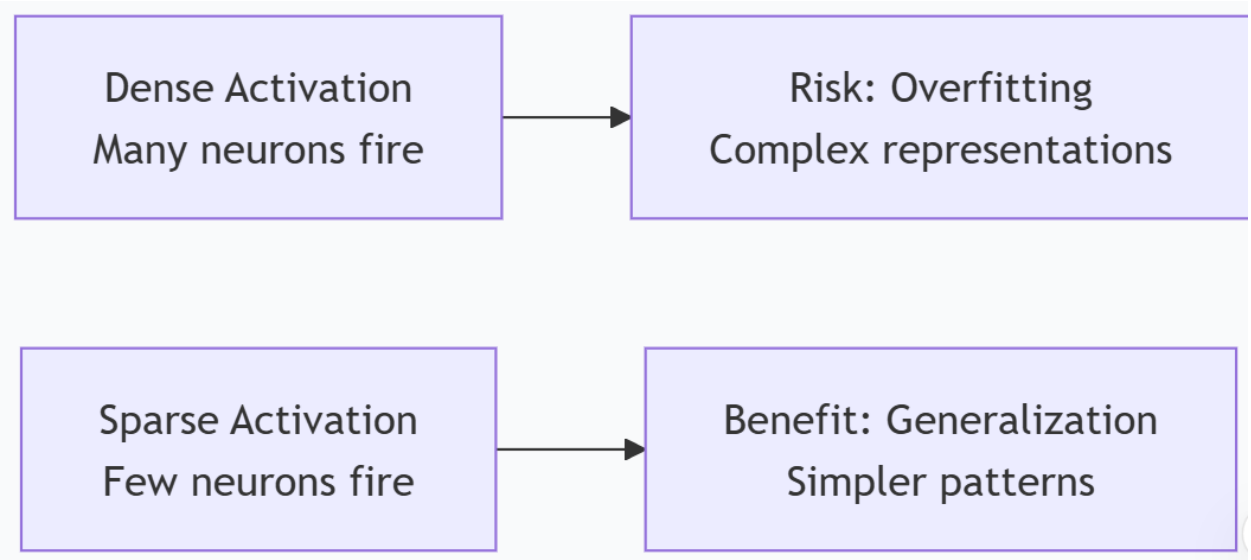
7.10 Sparse Representations

- **Definition**: Encourage model to activate only a few neurons at a time.
- **Example**: In autoencoders, only a few hidden units fire for each input.
- **Short Explanation**: Leads to simpler, interpretable, efficient representations.



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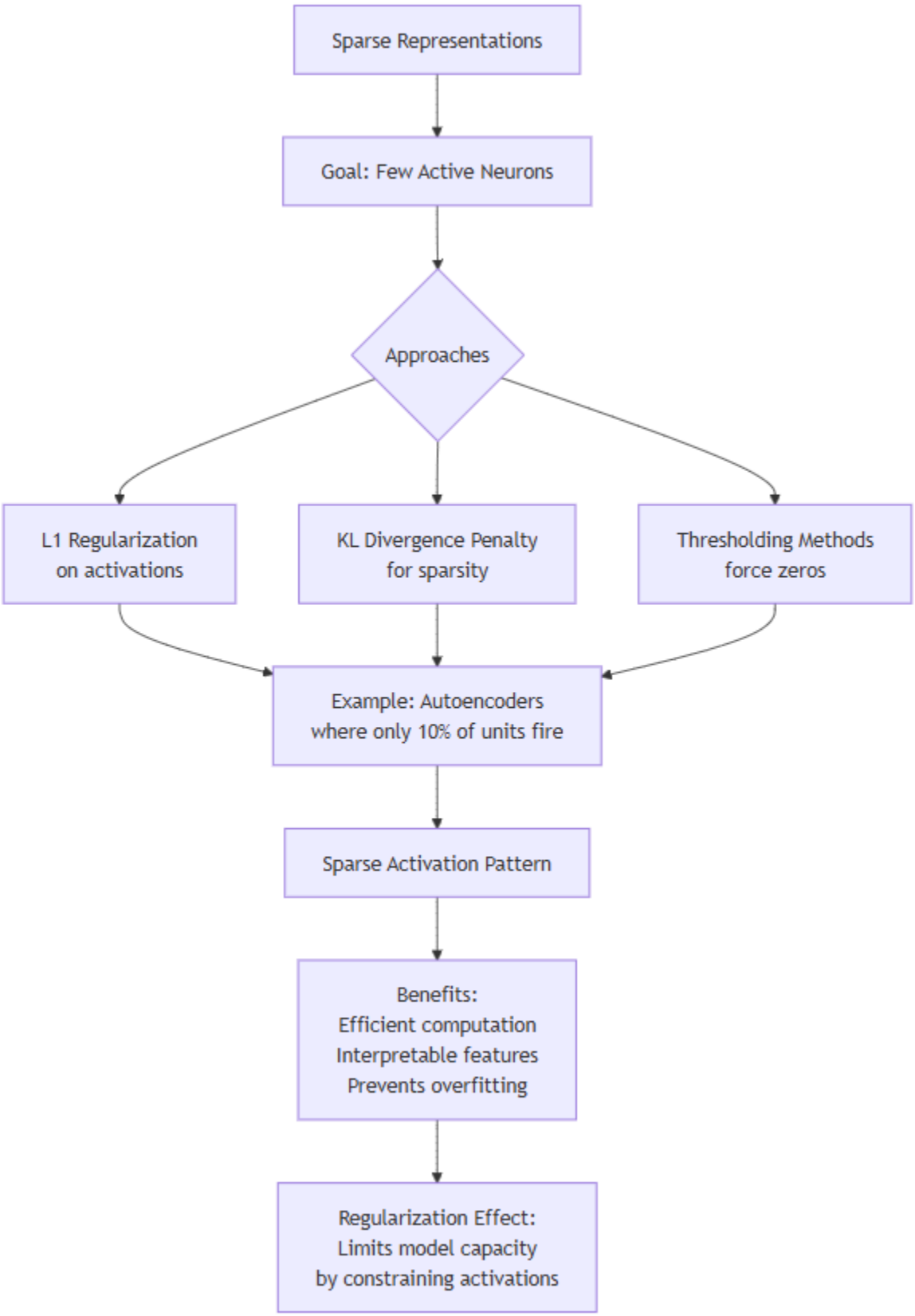

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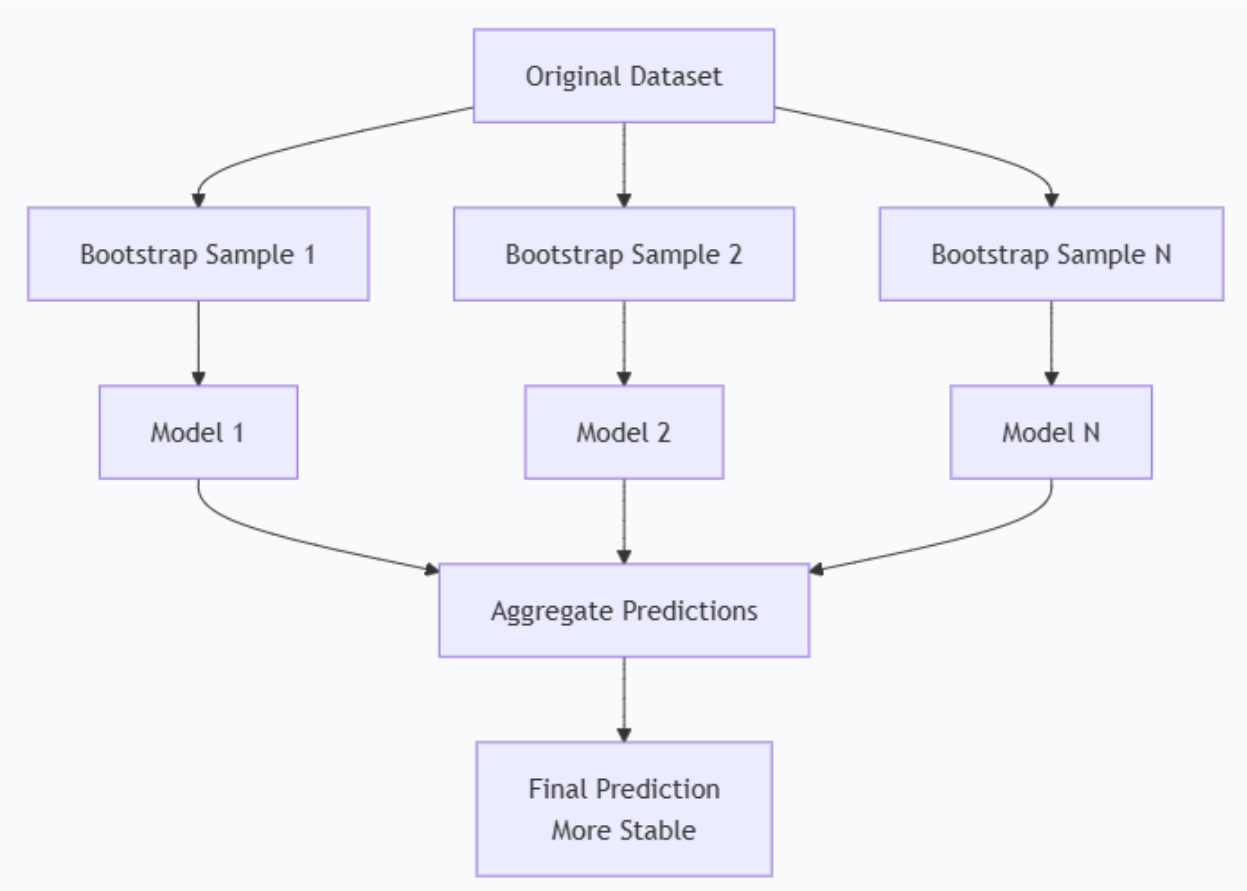
7.11 Bagging and Other Ensemble Methods

- **Definition**: Train multiple models and combine their predictions.
- **Example**: Random Forest (bagging with decision trees).
- **Short Explanation**: Reduces variance, smooths predictions, increases robustness.



7.12 Dropout

- **Definition**: Randomly drop (deactivate) neurons during training.
- **Example**: In each iteration, 30% of hidden units are ignored.
- **Short Explanation**: Prevents co-adaptation, makes network act like an ensemble.

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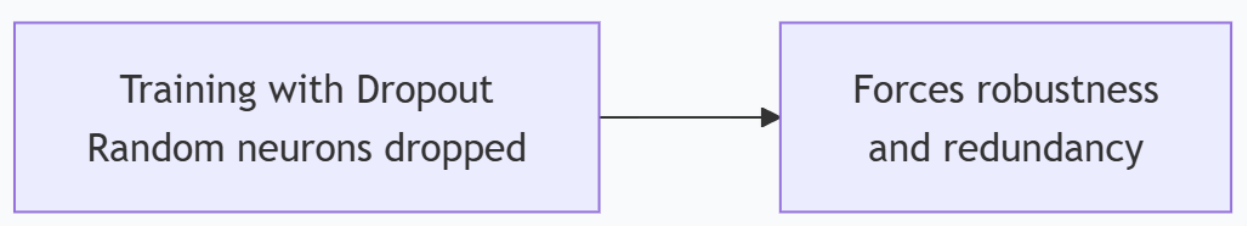
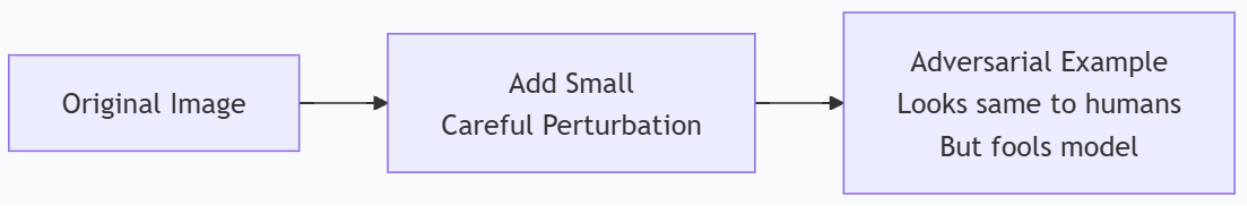


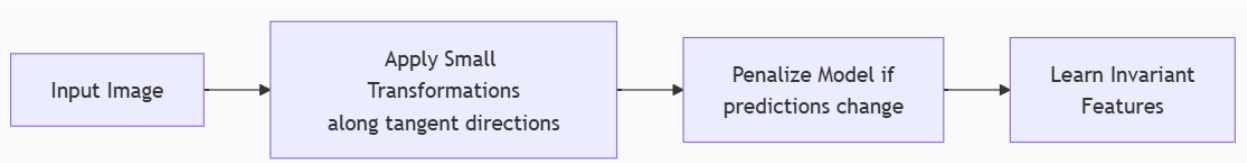
7.13 Adversarial Training

- **Definition**: Train model with adversarial examples (inputs with small but tricky perturbations).
- **Example**: Slightly altered image that fools classifier → include it in training.
- **Short Explanation**: Improves robustness against attacks and noise.



7.14 Tangent Distance, Tangent Prop, and Manifold Tangent Classifier

- **Definition**: Regularization methods that make models invariant to small transformations along a data manifold.
- **Example**: For digit "3," classifier should still recognize it if slightly rotated/skewed.
- **Short Explanation**: Ensures learning is guided by true underlying data structure, not irrelevant distortions.



1. Parameter-based Regularization (Keep weights simple, small, or shared) • Case: ImageNet Classification (AlexNet, 2012) • Method: CNNs with parameter sharing (filters reused across	Parameters → Simplicity & sharing	Parameters → AlexNet/LeNet (CNN filters = parameter sharing).

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pixels) • Impact: Made large-scale image recognition possible • Krizhevsky et al., 2012		
2. Data-based Regularization (Expose model to richer or trickier data) • Case: MNIST Digit Recognition (Elastic distortions, 1998–2003) • Method: Augmentation (rotations, shifts, stretching) • Impact: Robust to handwriting variations • Simard et al., 2003	Data → Augment & diversify	Data → MNIST (augmentation with distortions).
3. Training Process-based Regularization (Control overfitting during learning) • Case: ImageNet (AlexNet, 2012) • Method: Dropout (randomly drop neurons during training) • Impact: Reduced co-adaptation, improved generalization • Srivastava et al., 2014	Training → Stop overfitting in time	Training → AlexNet (dropout = landmark regularizer).
4. Multi-Model / Task-based Regularization (Learn from multiple tasks or models) • Case: Credit Card Fraud Detection (Kaggle) • Method: Bagging & Ensembles (XGBoost) • Impact: Reduced false alarms, robust anomaly detection • Chen & Guestrin, 2016	Multi-model → Combine wisdom of many models	Multi-model → Kaggle fraud/XGBoost (ensembles = state-of-the-art).

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